C-D-221G
February 29, 1972
SUPERSEDING
Fed. Spec. C-D-221F
July 3, 1968

FEDERAL SPECIFICATION

DESSERT POWDERS AND GELATIN, PLAIN, EDIBLE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers plain (unflavored) gelatin, flavored dessert powder prepared with gelatin, and flavored dessert powder prepared with starch.

1.2 Classification. Dessert powders and gelatin shall be of the following types as specified (see 6.1). Type II and III flavors shall be as specified (see 6.8).

Type I	-	Gelatin, plain (unflavored).
Type II	-	Dessert powder (flavored), prepared with gelatin.
Type III	-	Dessert powder (flavored), cooking type prepared with starch.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-C-110	-	Cellophane (Coated and Noncoated Regenerated Cellulose Film).
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-C-29	-	Canned Subsistence Items, Packaging and Packing Of.

Federal Standards:

FED-STD-101	-	Preservation, Packaging and Packing Materials: Test Procedures
FED-STD-123	-	Marking for Domestic Shipment (Civilian Agencies).

FSC 8940

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-A-148 - Aluminum Foil.
MIL-L-1497 - Labeling of Metal Cans for Subsistence Items.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-668 - Sanitary Standards for Food Plants.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

U.S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

U.S. Department of Agriculture

U.S. Standards for Condition of Food Containers

(Application for copies should be addressed to the Information Division, Consumer and Marketing Service, U.S. Department of Agriculture, Washington, D.C. 20250.)

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, P.O. Box 540, Benjamin Franklin Station, Washington, D.C. 20044.)

The United States Pharmacopoeial Convention, Inc.

Pharmacopoeia of the United States

(Application for copies should be addressed to the U.S. Pharmacopoeial Convention, Inc., 46 Park Ave., New York, New York 10016.)

American Public Health Association

Standard Methods for the Examination of Dairy Products

(Application for copies should be addressed to the American Public Health Association, 1740 Broadway, New York, New York 10019.)

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National Research Council

Food Chemical Codex

(Application for copies should be addressed to the Office of Director, National Academy of Science, National Research Council, Washington, D.C. 20418.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, **Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.**)

3. REQUIREMENTS

3.1 Material. The material shall be clean, sound, wholesome and free from foreign odors, insect infestation and any other foreign matter.

3.1.1 Gelatin. Gelatin shall be a high quality, ground gelatin. The moisture content shall be not more than 13.0 percent. In hot solution it shall have no foreign odor or taste, shall be clear, and shall be of light color. The product shall be free from grit, dirt, or other foreign matter. The gelatin shall be free from preservatives and bleaches. In addition, the gelatin shall have a standard plate count of not more than 5,000 per gram (g.) when tested as specified in 4.3.3.1, and shall be free from coliform organisms when tested as specified in 4.3.3.2.

3.1.2 Sweetening ingredients. Sweetening ingredients shall consist of white, refined cane or beet sugar or dextrose or any combination thereof.

3.1.3 Edible organic acids. Edible organic acids shall be anhydrous citric or other organic acids which meet the requirements of 3.5, and give an equivalent finished product (see 3.3.2).

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3.1.4 Flavors. Orange and lemon oils shall be of U.S. Pharmacopoeia quality and, if multiple fold, shall be made from cold pressed oils of U.S. Pharmacopoeia quality. Lime oil shall be distilled oil of lime, free from off odors or flavors. Non-oil flavors may be natural or artificial, or a combination of both. All flavors shall be protected from evaporation, oxidation, rancidity, or other types of deterioration by stabilizing them with vegetable gums or other edible materials which do not affect the appearance or utilization of the product and are readily soluble in water.

3.1.5 Coloring. Coloring shall be of vegetable origin or U.S. certified food color, or a combination of both.

3.1.6 Buffering salts. Buffering salts shall be citrate or phosphate salts, or a combination thereof and shall conform to Food Chemical Codex.

3.1.7 Starch. Starch shall be of high quality and bland taste. Any type of edible starch that will give the desired result will be acceptable.

3.1.8 Salt. Salt shall be refined sodium chloride with or without anti-caking agent. Iodized salt shall not be used.

3.1.9 Cocoa. Cocoa powder shall be a medium fat cocoa made from the nibs of domestically roasted, matured, well fermented cocoa beans which have been properly dried. Either the nibs or liquor from the nibs shall be mildly alkalized. The pH shall be not less than 6.0 nor more than 7.2. Chemically extracted cocoa, in whole or in part, is not acceptable.

3.2 Type I. Type I gelatin shall conform to the requirements of 3.1.1. The jelly strength shall be not less than 225 g. when tested as specified in 4.3.2.

3.3 Type II.

3.3.1 Formula. Type II, gelatin-dessert powder, shall be prepared from ingredients in the following proportions:

Ingredient	Percent by weight
Gelatin	Sufficient to produce the required finished jelly strength.
Sweetening ingredients 1/	To make 100 percent.

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Ingredient	Percent by weight
Edible organic acids	Sufficient to produce the required pH.
Salt 2/ not more than	0.5 percent.
Buffering salts 3/ not more than	The weight of the acid used and sufficient to maintain the required pH.
Flavor	Sufficient to produce the distinct characteristic flavor in the prepared gelled dessert, (see 3.3.2).
Color	Sufficient to produce the accepted color for the flavor required.

1/ If a mixture of sucrose and dextrose is used, the dextrose content shall not exceed 18 percent by weight of the finished product.

2/ This added salt does not include the salt or other chlorides contained in the other ingredients.

3/ The weight of buffering salts used shall be computed on an anhydrous basis.

3.3.2 Finished product. The finished product shall be composed of ingredients having approximately the same range of particle sizes, thoroughly mixed and blended into a free-flowing powder. Not less than 99 percent of mix shall pass through a U.S. Standard No. 30 sieve. Slight caking is permissible provided that the lumps caused by the caking will disperse readily and pass through the sieve specified when slight pressure is applied. The moisture content of the powder shall not exceed 2.0 percent. The total salt content of the powder (calculated as NaCl), shall not exceed 1.0 percent. The jelly strength of the gelled dessert shall be not less than 60 g. when tested as specified in 4.3.2. (The texture of the gelled dessert shall be firm and tender, not soft, runny, tough, or rubbery when the dessert is prepared from the powder as specified in 4.3.1.2.2). The gelled dessert shall have a pH of not less than 3.0 nor more than 4.5. The flavor of the prepared dessert shall be of such quality and intensity that the flavor can be readily identified as that of the fruit it represents or imitates. The color of the prepared dessert shall be characteristic of the fruit which the flavor represents or imitates.

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3.4 Type III.

3.4.1 Formula. Type III, starch-dessert powder, shall be prepared from ingredients in the following proportions:

Ingredient	Percent by weight
Starch not more than 1/	30 (butterscotch and chocolate flavors). 42 (lemon flavor). 35 (all other flavors).
Sweetening ingredients 2/	To make 100 percent.
Flavor 3/	Sufficient to produce the distinct characteristic flavor in the prepared product.
Color	Sufficient to produce the accepted color in the prepared dessert for the flavor specified.

1/ Wheat flour may be used as a substitute for starch, but, when used, wheat flour shall not exceed 10 percent by weight of the finished product.

2/ When a mixture of sucrose and dextrose is used, the dextrose content shall not exceed 40 percent by weight of the finished product.

3/ When chocolate-flavored dessert powder is specified, the cocoa content of the product shall be not less than 12.5 percent by weight.

NOTE: Optional ingredients, such as gelatin, dried eggs (certified as salmonella negative by USDA), tapioca granules, salt, acids and stabilizers, may be used provided they do not affect the quality of the finished product.

3.4.2 Finished product. The ingredients shall be thoroughly mixed and blended into a free-flowing powder. The moisture content shall not exceed 3.5 percent, and the powder shall be free from grit, dirt, or other foreign matter. The dessert, prepared from type III powder as specified in 4.3.1.2.3 shall be smooth-textured, tender, and of such consistency that it has the creamy, wholesome, rich, appetizing appearance and eating quality characteristics of this type of dessert. The prepared dessert shall be free from froth, air bubbles, lumpiness, and graininess, and shall not be rubbery, tough, or watery. The flavor of the finished dessert shall be distinct, wholesome, full-bodied, and appetizing, and it shall be free from foreign or objectionable tastes and odors.

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3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.6 Workmanship. The product shall be prepared, processed and packaged under modern sanitary conditions and in accordance with good commercial practice. For military agencies, the product shall be processed in establishments meeting the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling. Samples of finished products covered by this specification, and samples of any materials, components, or constituents entering into the preparation of the finished products may be taken at any time by the Government inspector to determine compliance with the requirements.

4.2 Inspection.

4.2.1 General. Unless otherwise specified in the contract, inspection shall be made by the procuring agency or a duly authorized representative, at the time and place designated by the procuring agency. It may be made at the site of manufacture both during and after processing, at any suitable point in transit, or after delivery to the point of destination. The passing as satisfactory of any detail of processing or materials shall not relieve the supplier of responsibility for faulty workmanship or materials, which may be discovered at any time prior to final acceptance.

4.2.2 Final inspection. Unless otherwise specified, final inspection shall be made after delivery to the point of destination (see 6.4).

4.2.3 Condition inspection of containers. When inspection of product for quality and condition is performed by the U.S. Department of Agriculture, the United States Standards for Condition of Food Containers shall apply. (For applicable criteria governing inspection of containers for the Military, see 4.2.6.3).

4.2.4 Responsibility for inspection. The supplier is responsible for meeting all the provisions of the specification prior to submission of supplies to the Government for acceptance. Unless otherwise specified in the contract or order, the supplier is required to perform the examination and tests prescribed herein, using his own or any other inspection facilities and services acceptable to the Government.

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4.2.4.1 Inspection records of the examination and tests shall be complete and made available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2.5 Preacceptance inspection.

4.2.5.1 Military agencies. Product shall be rejected if produced in plants not meeting the requirements of MIL-STD-668.

4.2.5.2 Civil agencies. When specified (see 6.1), inspection of plants producing the product will be conducted upon the request of the contracting officer, to the appropriate Government inspection agency. Award of contracts will be limited to establishments known to maintain proper sanitary conditions or which have received prior sanitary approval recommended by the inspection agency.

4.2.6 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except as otherwise indicated hereinafter.

4.2.6.1 Component and material inspection. In accordance with 4.2.4, components and materials shall be inspected in accordance with the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.6.1.1 Examination of ingredients. Conformance of each ingredient to identity and condition requirements shall be determined in accordance with the examination criteria contained in 4.2.6.1.1.1 and 4.2.6.1.1.2. One or more nonconformances shall indicate an unacceptable ingredient, and use of such shall be cause for rejection of the involved finished product.

4.2.6.1.1.1 Identity. Conformance to identity requirements (kind, quality, standard, method of manufacture, certification of salmonella negative egg product - as applicable) shall be ascertained by examination of pertinent labels, invoices or other valid documents.

4.2.6.1.1.2 Condition. Each ingredient shall be examined organoleptically as necessary to determine conformance to condition requirements of 3.1 and as may be contained in the pertinent requirement paragraph. The sample unit for examination shall be approximately 8 ounces of product from a container (coloring and flavoring ingredients - approximately 1/4 ounce). The sample size shall be five. In the event a lot consists of less than five containers, a sample unit shall be extracted from each container in the lot.

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4.2.6.1.2 Testing of components. Testing of components described in this specification, and changes to the testing of components listed in subsidiary specifications shall be as shown in table I. Results shall apply to lot average requirements. Nonconformance to test requirements shall indicate an unacceptable component, and use of such shall be cause for rejection of the involved quantity of finished product.

TABLE I. Testing of components

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref. and rqmt. para.
Waxed paper	2 sq. yds.	Rolls or wrappers	S-1	Basis weight of paper	Nearest lb./ream	5.1.2.1 4.3
				Weight of wax	Nearest lb./ream	
Laminated foil	2 sq. yds.	Rolls or wrappers	S-1	Thickness of laminated foil 1/	Nearest 0.00001 inch	5.1.2.1 4.3
				Basis weight of sulfite tissue	Nearest lb./ream	
				Weight of wax	Nearest lb./ream	
Bag (laminated glassine)	5 bags	Bags	S-1	Basis weight (each ply)	Nearest lb./ream	5.1.2.2 4.3
				Weight of wax or wax-resin (as applicable)	Nearest lb./ream	
Gelatin	1 lb. composite 2/	Pounds	S-2	Moisture	Nearest 0.1 percent	3.1.1 4.3.2
				Standard plate count	Nearest 100 bacteria per gm.	4.3.3.1
				Coliform	Pass or fail	4.3.3.2

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TABLE I. Testing of components (cont'd)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref. and rqmt. para.
Cocoa	1 lb. composite 2/	Pounds	S-2	pH	Nearest 0.1 pH	3.1.9 4.3

1/ Conformance to this requirement may be ascertained by a review of labels, invoices or other valid documents.

2/ The composite shall be derived from the number of containers indicated by the inspection level.

4.2.6.2 Process examination. Unless otherwise specified, (see 6.1) examination shall be performed to determine conformance to formulation requirements of 3.3.1 and 3.4.1. Formulation records shall be maintained. Nonconformance shall be cause for rejection of the involved finished product.

4.2.6.3 Examination of end item. The end item shall be examined in accordance with table III through V and 4.2.6.3.1 through 4.2.6.3.7. The sampling plans for the aforementioned tables shall be in accordance with table II. The acceptable quality levels (AQLs) shall be expressed as defects per hundred units for tables III and V, and percent defective for table IV.

TABLE II. Sampling plans

Table	Sample unit	Lot size expressed in	Inspection level	AQLs Major	Minor	Total
III	1 filled and closed package 1/	Primary container	I	1.5	--	4.0
IV	1 filled primary container	Primary container	S-3	--	2.5	--
V	1 filled primary container	Primary container	S-2	1.0	--	4.0

1/ PPP-C-29 cans shall not be included in this examination.

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TABLE III. External examination of package

Category		Defect
Major	Minor	
101		Container not style specified or fabricated from kinds of material specified.
	201	Carton inclosing bag not closed.
	202	Tear, hole or open seal in outer wrap of carton.
102		Tear or hole completely through container, sifter, improper closure (can or canister), open seal or closure (bag).
	203	Bag not closed or sealed by method specified.
	204	Bag not glued to two sides of carton, or double crimp folded on top.
103		Not clean (other than cans).
	205	Can has severe dent.
104		Pitted rust. 1/
	206	Can not clean. 2/
105		Nomenclature or directions for use missing, incorrect or illegible. 3/
	207	Other required label information missing, incorrect or illegible. 3/

1/ Rust that can be wiped off with a soft cloth shall not be considered pitted rust.

2/ Cans showing a very thin film of grease which is discernible to the touch but not readily discernible visually are considered to be clean.

3/ Not applicable to cans.

TABLE IV. Examination of net weight 1/ 2/

Category	Defect
Minor	
201	More than 5 percent under specified net weight. (for containers having one pound or less).
202	More than 2 percent under specified net weight. (for containers having more than one pound).

1/ Report results to nearest gram for container required to have a net weight of 4-1/2 ounces or less; nearest 1/16 ounce for containers required to have more than 4-1/2 ounces but not more than 1 pound, and nearest 1/4 ounce for containers required to have more than 1 pound.

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2/ Lot shall be rejected if sample data indicate a lot average net weight less than specified net weight.

TABLE V. Examination of contents of primary container 1/

Categories		Defects
Major	Minor	
101		Directions for use sheet in cans missing (when information is not on can label).
	201	Directions for use sheet not placed on top of product in can (when information is not on can label).
102		Required information missing, incorrect or illegible.
	202	Size of lettering not as required.
		Dry product
	203	Not free flowing.
	204	Not thoroughly mixed (types II and III).
		Prepared product 2/
103		Type II product soft, runny, tough or rubbery.
104		Type III product not smooth and creamy (i.e., tough, rubbery, grainy, lumpy, watery).
	205	Type III product not free of froth or air bubbles.
105		Flavor of type II or III product not typical of the flavor being procured.
106		Flavor of type I product not basically bland (foreign flavor).
107		Foreign or objectionable odor. (Applicable to type I, II or III).
108		Color not typical of flavor being procured.
109		Type I product not clear or light in color.

1/ Presence of foreign material (e.g., metal, wood, filth, glass, insects, paints, etc. shall be cause for rejection of the lot).

2/ Prepare product in accordance with 4.3.1.2.

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4.2.6.3.1 Examination of particle size for type II product. The sample shall be a 100-gram composite of product derived from the number of primary containers indicated by inspection level S-2. Lot size shall be expressed in primary containers. Procedures shall be in accordance with 4.3.1.1. Result shall be reported to the nearest percent. Less than 99 percent by weight through the U.S. Standard No. 30 sieve shall be cause for rejection of the lot.

4.2.6.3.2 Examination of cans. When packaging is required to be in accordance with PPP-C-29, examination of cans for condition, external can coating (when required) and seam leakage shall be in accordance with the applicable quality assurance provisions of that specification.

4.2.6.3.3 Examination of can label. Examination shall be in accordance with the quality assurance provisions of MIL-L-1497.

4.2.6.3.4 Examination of waterproof case liners. Examination of filled and closed case liners shall be in accordance with quality assurance provisions in the appendix of MIL-L-10547.

4.2.6.3.5 Examination of packing and shipping containers (for cans). Examination shall be in accordance with the applicable quality assurance provisions in PPP-C-29.

4.2.6.3.6 Examination of shipping containers (for cartons and canisters). Examination of the filled and closed shipping containers shall be in accordance with the quality assurance provisions in the appendix to PPP-B-636. In addition, the following defects shall be included in the examination: (1) Major - Protective pads and tape (when used in lieu of corner pad) for level A and B pack missing, not as specified, or not completely covering stitches; Minor - Net weight marking on case indicates level A or B pack cartons exceeds 48 pounds; (2) Major - Markings missing, incorrect or illegible. For level C pack as specified in 5.2.2.3.1 only the defects applicable to number and variety of flavors, and markings shall apply. For level C pack as specified in 5.2.3, only the defect applicable to marking shall apply.

4.2.6.3.7 Examination of unit loads. Unit loads shall be examined in accordance with the quality assurance provisions of MIL-L-35078.

4.2.6.4 Testing of finished product. The finished product shall be tested for moisture, jelly strength, salt, and pH, as applicable to the specific type of product. Test procedures shall be in accordance with 4.3. The sample for testing shall be a 1-pound composite derived from the number

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of primary containers indicated by inspection level S-2. Lot size shall be expressed in terms of the primary containers. Test results shall be reported to the same unit or decimal place as specified for the pertinent requirement. A test failure shall be cause for rejection of the lot.

4.3 Test methods. Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum such quality assurance, the supplier will use the procedures set forth herein. In case of dispute as to examination or test results, the procedures specified herein will govern.

4.3.1 Examination procedures.

4.3.1.1 Sieve. Place a 100-gram sample in the specified sieve, attach cover and catch pan. With the left hand, hold the sieve assembly at an angle of about 30 degrees from the horizontal so that the upper edge of the tilted plane is towards the right hand. Shake by striking the right side of the sieve assembly with a short sharp stroke with the right hand moving in a horizontal plane. Strike at the rate of about 150 times per minute. After each 25 strokes, turn the sieve assembly 1/6 of a revolution in the same direction. Continue shaking for 2 minutes. Weigh and calculate the percentage of the material which passes through the sieve. (NOTE: Sieves may close up because of moisture and oil conditions. In cases such as these, a hairbrush may be used to break the bridging effect).

4.3.1.2 Examination for flavor, texture, and consistency. (No alternate procedure permitted).

4.3.1.2.1 For type I, 1 oz. of type I Gelatin shall be dissolved in 1 pint of boiling water and subsequently cooled to room temperature to determine conformance to 3.1.1.

4.3.1.2.2 For type II, three ounces of type II powder shall be dissolved in 1 pint of hot distilled water, subsequently cooled to 45°F., and held at that temperature for not more than 4 hours. The formed gelatin shall then be examined to determine conformance to 3.3.2.

* 4.3.1.2.3 For type III, one part (butterscotch or chocolate flavor), or 3/4 parts (all other flavors) by weight of the type III powder shall be mixed with 4 parts by weight of reconstituted nonfat dry milk. The mixture shall be cooked over low heat with constant stirring to avoid scorching or undue sticking, until thickened, and then poured into individual molds or a shallow pan and

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refrigerated at a temperature of approximately 45°F. for 3 hours. It shall be allowed to stand at room temperature for 30 minutes. The prepared pudding shall then be examined to determine conformance to 3.4.2.

4.3.2 Test procedures.

4.3.2.1 Chemical analyses. Unless otherwise specified, chemical analyses shall be conducted in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists as indicated below:

Test	Source	Method
Cocoa:		
pH	Chapter: Cereal foods	Hydrogen-ion concentration electrometric method
Gelatin:		
Moisture	Chapter: Cereal foods	Vacuum oven
Finished product:		
Moisture	Chapter: Sugars and sugar products, Section: Sugars, sirups and molasses	Moisture - vacuum drying
Salt	Chapter: Egg and egg products	Chlorine method II
Jelly strength	Gelatin, dessert preparations and mixes	Jelly strength
Dextrose (Type II)	Chapter: Gelatin, dessert preparations and mixes	Glucose, Official Final Action
Dextrose (Type III)		Sucrose and Glucose Final Action

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4.3.2.2 For type II. The pH shall be determined at 25°C. using a glass-calomel electrode system and a potentiometer. The sample shall be the same gelled sample as that used for the jelly strength determination after adjusting the temperature of the sample to 25°C.

4.3.3 Bacteriological analyses. Unless otherwise specified, bacteriological examinations shall be made in accordance with Standard Methods for the Examination of Dairy Products. In the preparation of a sample for examination, 5 g. shall be weighed out aseptically into a 95-ml. sterile water blank containing glass beads held at 45°C. and the resulting 5 percent mixture (equivalent to a 1:20 dilution) tempered in a water bath at 45°C. with intermittent shaking for not more than 15 minutes. A new 5 percent solution shall be made for each determination.

4.3.3.1 Standard plate count. Bacterial estimates by the standard plate count shall be made in triplicate using 1 ml. and 0.1 ml. portions of the 5 percent solution for plating with tryptone glucose extract agar. Incubation shall be at 32°C. for 72 hours. The results shall be expressed as the standard plate count per gram based on the average bacterial count of 3 plates in the significant dilution.

4.3.3.2 Coliform organisms. A presumptive test for coliform organisms shall be made using 1 ml. of the 5 percent solution in each of 3 tubes of brilliant green bile lactose broth. Incubate tubes at 35°C. for 48 hours. Gas formation within 48 hours constitutes a positive presumptive test.

4.3.4 Basis weight of paper. Paper shall be tested for basis weight in accordance with Method 5022 of FED-STD-101.

4.3.5 Thickness of foil. Thickness shall be determined in accordance with MIL-A-148.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A, B, or C, as specified (see 6.1).

5.1.1 Level A. A net weight of 8 ounces or 24 ounces of the product, as specified (see 6.1), shall be filled into a size 211 by 400, or size 401 by 411 can respectively, in accordance with the level A requirements of PPP-C-29.

5.1.2 Level B. Eight ounces or 5 pounds of the type I product, 3 ounces or 24 ounces of the type II product, 3 to 4-1/2 ounces or 24 ounces or 5 pounds of the type III product, as specified (see 6.1), shall be packaged in accordance with 5.1.2.1 or 5.1.2.2, as specified (see 6.1).

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5.1.2.1 Overwrapped carton. The product shall be packaged in a folding carton which shall be sealed and then overwrapped in one of the following materials:

(a) A sheet of aluminum foil, of not less than nominal 0.00035-inch thickness, laminated with 8 pounds per ream ± 7-1/2 percent of wax to a sulfite tissue having a nominal basis weight of 12 pounds (24 by 36 - 500) per ream ± 5 percent. The tissue is laminated with 16 pounds per ream ± 7-1/2 percent of wax to a sulfite tissue having a nominal basis weight of 8 pounds per ream ± 5 percent.

(b) A sheet of sulfite paper having a nominal basis weight of 23 pounds per ream ± 5 percent (24 by 36 - 500) wet waxed overall with 14 pounds per ream ± 7-1/2 percent of wax.

(c) A sheet of cellophane conforming to type IV, class 2, of L-C-110.

5.1.2.2 Bag in carton. The product shall be packaged in a bag fabricated from type IV, class 2, cellophane conforming to L-C-110 and enclosed in a paperboard folding carton. The bag shall be closed by heat sealing. Alternatively, the bag may be constructed of two sheets of glassine, each having a nominal basis weight of 20 pounds per ream ± 5 percent (24 by 36 - 500) and laminated to each other with 7 pounds per ream ± 7-1/2 percent of micro crystalline wax-resin blend or amorphous wax. The bag shall be glued to the carton on two sides and shall be double crimp folded on top. Alternatively, the glassine bag shall be coated on the inside with polyethylene and shall be closed by heat sealing.

5.1.3 Level C. The product shall be packaged in a metal can, carton, heat sealed moisture-resistant bag, or canister to protect it from deterioration or damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements. When specified (see 6.1), the product, in the amounts and can sizes specified in 5.1.1, shall be packaged in accordance with the level C requirements of PPP-C-29.

5.2 Packing. The product shall be packed in accordance with 5.2.1, 5.2.2 or 5.2.3, as applicable.

5.2.1 Cans. The product in cans shall be packed in accordance with the level A, B, or C requirements of PPP-C-29, as specified (see 6.1 and 6.5).

5.2.2 Cartons. The product in cartons shall be packed in accordance with level A, B, or C as specified (see 6.1). Not more than 24 8-ounce cartons of type I product, 48 3-ounce cartons of type II product, 24 3 to 4-1/2-ounce cartons of type III product, 12 24-ounce cartons of type II or III product or not more than 8 5-pound cartons of type I or type III product shall be packed in a fiberboard box normally used for the product, and not more than 48 pounds of the product shall be overpacked in a snug-fitting shipping container in accordance with level A or B.

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5.2.2.1 Level A. The shipping container shall be a snug-fitting fiberboard box, constructed, closed, and reinforced in accordance with style RSC, V2s, of PPP-B-636. The shipping case shall be furnished with a waterproof case liner, fabricated and closed in accordance with MIL-L-10547 for subsistence items. If the flaps of the shipping container are closed by stitching, the case liner shall be protected by a chipboard pad between the stitches and the case liner. The case liner shall be protected from stitches in the manufacturer's pad with a chipboard corner joint or a strip of pressure-sensitive tape placed over the stitches.

5.2.2.2 Level B. The shipping container shall be a fiberboard box, constructed, closed, and reinforced in accordance with style RSC, V3c, V3s, or V4s, of PPP-B-636. The shipping container shall be furnished with a waterproof case liner as specified in 5.2.2.1. The case liner shall be protected from stitches as specified in 5.2.2.1.

5.2.2.2.1 When specified (see 6.1 and 6.6), the shipping container specified in 5.2.2.2 shall be reinforced with pressure-sensitive adhesive, filament-reinforced tape in accordance with the appendix of PPP-B-636.

5.2.2.3 Level C. Twenty-four 3-ounce cartons (8 of each specified flavor), 12-24 ounce cartons of type II and III, or six 5-pound cartons shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC, type CF or SF, class domestic, method II closure, of PPP-B-636.

5.2.2.3.1 When specified (see 6.1 and 6.7), the product in cartons shall be packed in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.2.3 Canisters. The product in canisters shall be packed in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.4 Labeling and marking.

5.4.1 Civil agencies.

5.4.1.1 Unit containers. Any commercial labeling that complies with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder will be acceptable. Directions for use shall be imprinted on each package.

5.4.1.2 Shipping containers. Shipping containers shall be marked in accordance with FED-STD-123.

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5.4.2 Military agencies.

5.4.2.1 Metal cans. Metal cans shall be labeled in accordance with MIL-L-1497 and shall include directions for use furnished by the manufacturer, except that for 24-ounce cans, directions for use shall be as specified in 5.4.2.3 or 5.4.2.4 and shall be lithographed on the body or end of can or printed on a sheet of good quality paper and placed on top of the product in each can. The printed sheet shall be nontoxic and shall impart no odor or flavor to the product.

5.4.2.2 Cartons and canisters. Cartons and canisters shall be labeled in accordance with 5.4.1.1 with directions for use furnished by the manufacturer except that 24-ounce containers shall be imprinted with directions for use as specified in 5.4.2.3 or 5.4.2.4.

5.4.2.3 Directions for Type II (24-ounce containers).

DIRECTIONS FOR USE

For approximately 35 servings. Measure 2 quarts boiling water into container, add contents of package. Stir until thoroughly dissolved, then add 2 quarts of cold or ice water. Pour into shallow pans and place in refrigerator or on crushed ice until set, about 4 hours. When adding fruit, replace part of the cold water with an equal quantity of fruit juice drained from the can, then add the drained fruit to the dessert just after it begins to thicken.

5.4.2.4 Directions for Type III (24-ounce containers).

DIRECTIONS FOR USE

Type III (vanilla and other flavors except butterscotch and chocolate)

For thirty-three (1/2 cup) servings

1. Mix contents (24 oz) and 13 oz avoirdupois (2-7/8 cups) nonfat dry milk.
2. Add 1 qt cool water. Stir until completely smooth.
3. Stir mixture into 2-3/4 qts near boiling water to which 5-1/3 oz (11 tbsp) butter or margarine has been added.
4. Heat slowly, stirring constantly, until pudding thickens and starts to boil.
5. Pour into shallow pans; cover and cool.

For pie, reduce hot water in step 3 to 2-1/2 qts.

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DIRECTIONS FOR USE

Types III (Butterscotch and Chocolate)

For twenty-five (1/2 cup) servings.

1. Mix contents (24 oz) and 10 oz avoirdupois (2-1/4 cups) nonfat dry milk.
2. Add 3 cups cool water. Stir until completely smooth.
3. Stir mixture into 2 quarts near boiling water to which 4 oz (1/2 cup) butter or margarine has been added.
4. Heat slowly, stirring constantly, until pudding thickens and starts to boil.
5. Pour into shallow pans; cover and cool.

For pie, reduce hot water in step 3 to 1-3/4 qts.

5.4.2.2 Fiberboard boxes (see 5.2.2). Fiberboard boxes shall be marked in accordance with MIL-STD-129.

5.4.2.3 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.2.4 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Type of product, flavors, and unit size of the container required (see 1.2 and section 5).
(b) When plant sanitation inspection is required (see 4.2.5.2).
(c) When process examination is required (see 4.2.6.2).
(d) Applicable levels of packaging and packing (see 5.1 and 5.2).
(e) Net weight and can sizes required for levels A and C packaging (see 5.1.1 and 5.1.3).
(f) When product for level C is to be packaged in accordance with PPP-C-29 (see 5.1.3).
(g) When packing in accordance with 5.2.2.2.1 or 5.2.2.3.1 is required.
(h) Type and class of unit load when unit loading is required (see 5.3).

6.2 If samples with bids are necessary, they should be specifically asked for in the invitation for bids, or request for proposal, and the particular purpose to be served by the bid sample should be definitely stated.

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6.3 Supplier's inspection. The supplier should be required to assure that the product conforms with the specification prior to submission to the Government for final inspection.

6.4 Destination inspection (for civil agencies). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for identity, condition and quantity only.

6.5 (For military agencies). In accordance with conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15.

6.6 Packing specified in 5.2.2.2.1 is intended for transfer at sea operations or specific overseas operations.

6.7 Packing specified in 5.2.2.3.1 is intended for direct shipment from the supply source to the first receiving activity for immediate use within CONUS.

6.8 Examples of flavors most commonly used are:

Dessert powder with gelatin (Type II)	Dessert powder with starch (Type III)
Cherry	Almond
Lemon	Butterscotch
Lime	Chocolate
Orange	Lemon
Raspberry	Orange
Strawberry	Vanilla

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

CIVIL AGENCY INTEREST:

GSA
AGR
VA
HEW

Project No. 8940-0255

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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